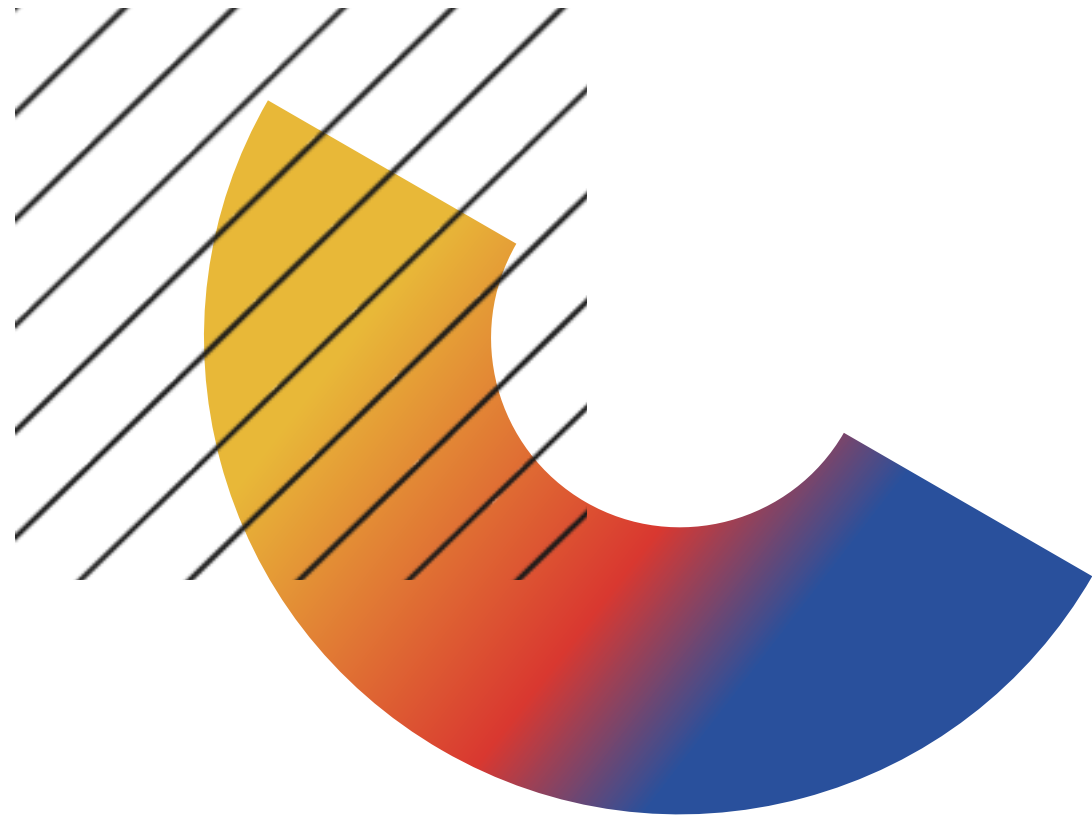


# Maximizing Revenue For Drivers

Through Payments Method

# Agenda



- Problem Statement
- Analysis & Findings
- Research Questions
- Hypothesis Testing
- Data Overview
- Recommendations
- Methodology

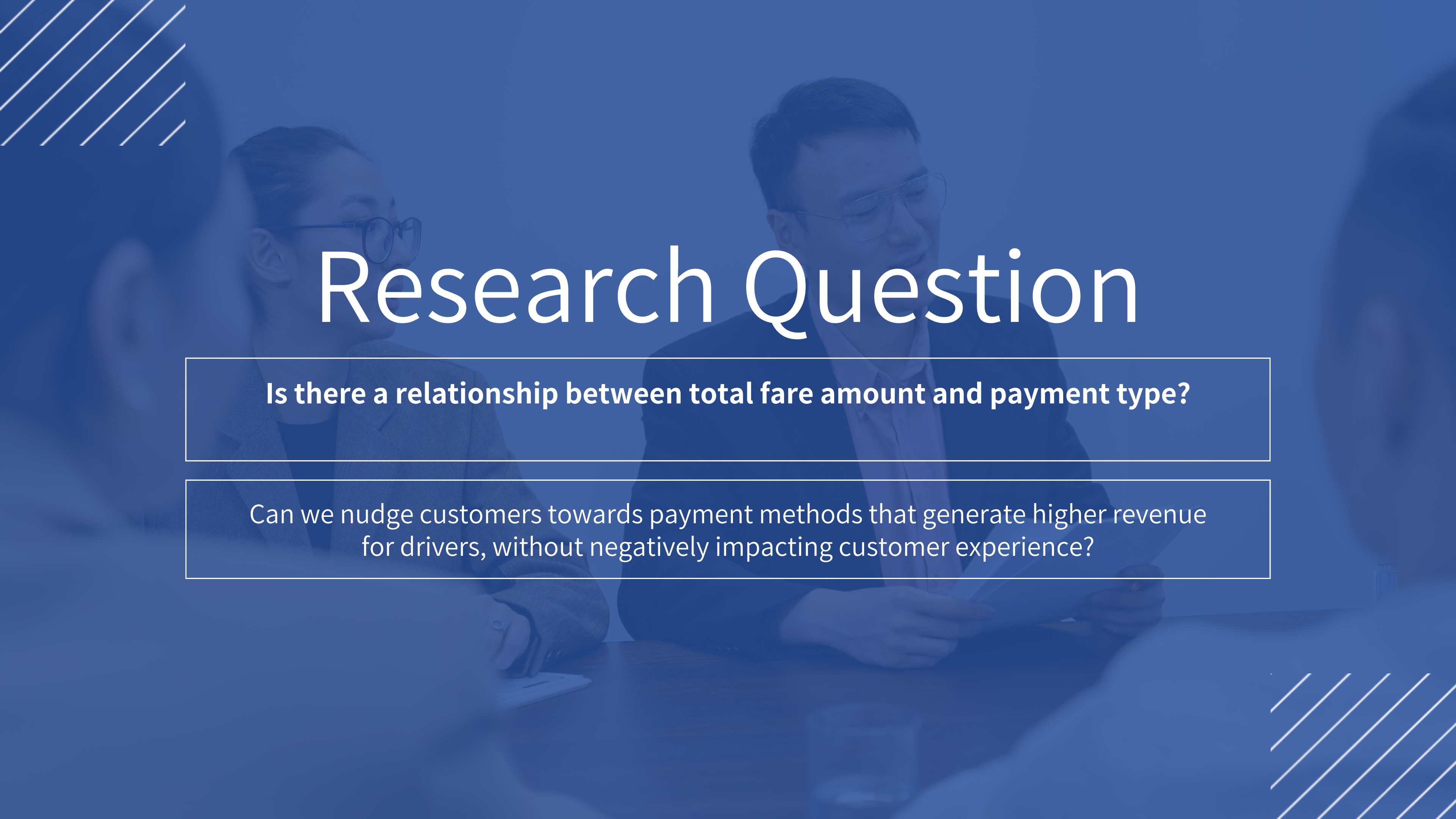
# Problem Statement

In the fast-paced taxi booking sector, making the most of revenue is essential for long-term success and driver happiness.

Our goal is to use data-driven insights to maximise revenue streams for taxi drivers in order to meet this need.

Our research aims to determine whether payment methods have an impact on fare pricing by focusing on the relationship between payment type and fare amount..





# Research Question

**Is there a relationship between total fare amount and payment type?**

Can we nudge customers towards payment methods that generate higher revenue for drivers, without negatively impacting customer experience?





# Data Overview

For this analysis, we utilized the comprehensive dataset of NYC Taxi Trip records, used data cleaning and feature engineering procedures to concentrate solely on the relevant columns essential for our investigation.

- `payment_type` (Credit Card / Cash)
- `passenger_count`(1 to 5)
- `fare_amount`
- `trip_distance`
- `duration`

|       | <code>passenger_count</code> | <code>payment_type</code> | <code>fare_amount</code> | <code>trip_distance</code> | <code>duration</code> |
|-------|------------------------------|---------------------------|--------------------------|----------------------------|-----------------------|
| 0     | 1                            | 1                         | 9.0                      | 2.50                       | 7.916667              |
| 1     | 1                            | 1                         | 11.0                     | 2.90                       | 11.100000             |
| 2     | 2                            | 1                         | 54.5                     | 19.98                      | 31.100000             |
| 3     | 3                            | 1                         | 31.5                     | 10.78                      | 0.000000              |
| 4     | 5                            | 1                         | 98.0                     | 30.43                      | 0.000000              |
| ...   | ...                          | ...                       | ...                      | ...                        | ...                   |
| 99995 | 1                            | 2                         | 5.0                      | 0.50                       | 5.083333              |
| 99996 | 1                            | 1                         | 14.0                     | 3.40                       | 15.516667             |
| 99997 | 1                            | 1                         | 29.0                     | 9.70                       | 20.216667             |
| 99998 | 1                            | 1                         | 5.5                      | 0.92                       | 4.983333              |
| 99999 | 1                            | 2                         | 6.0                      | 1.00                       | 4.816667              |

93680 rows x 5 columns



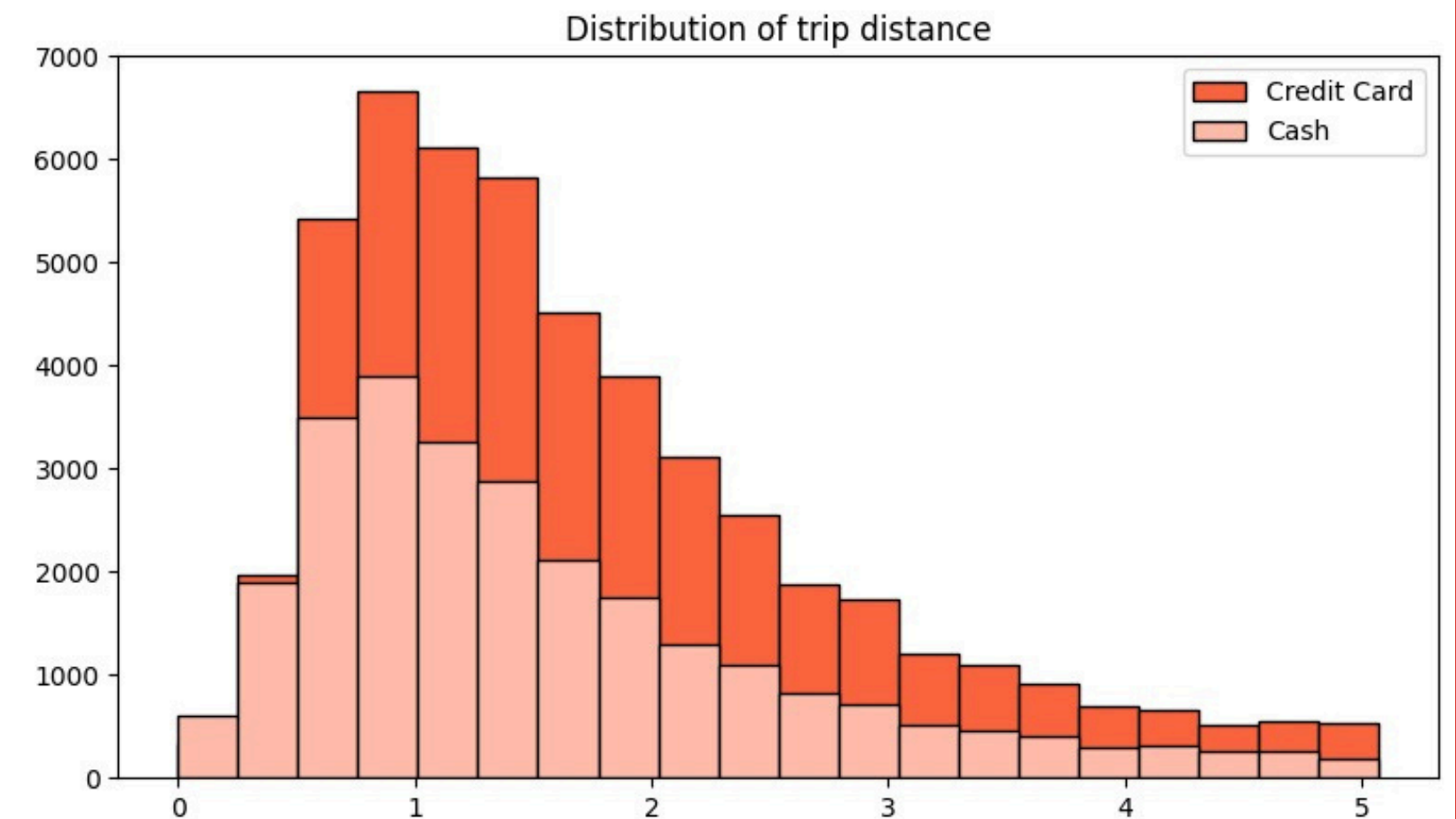
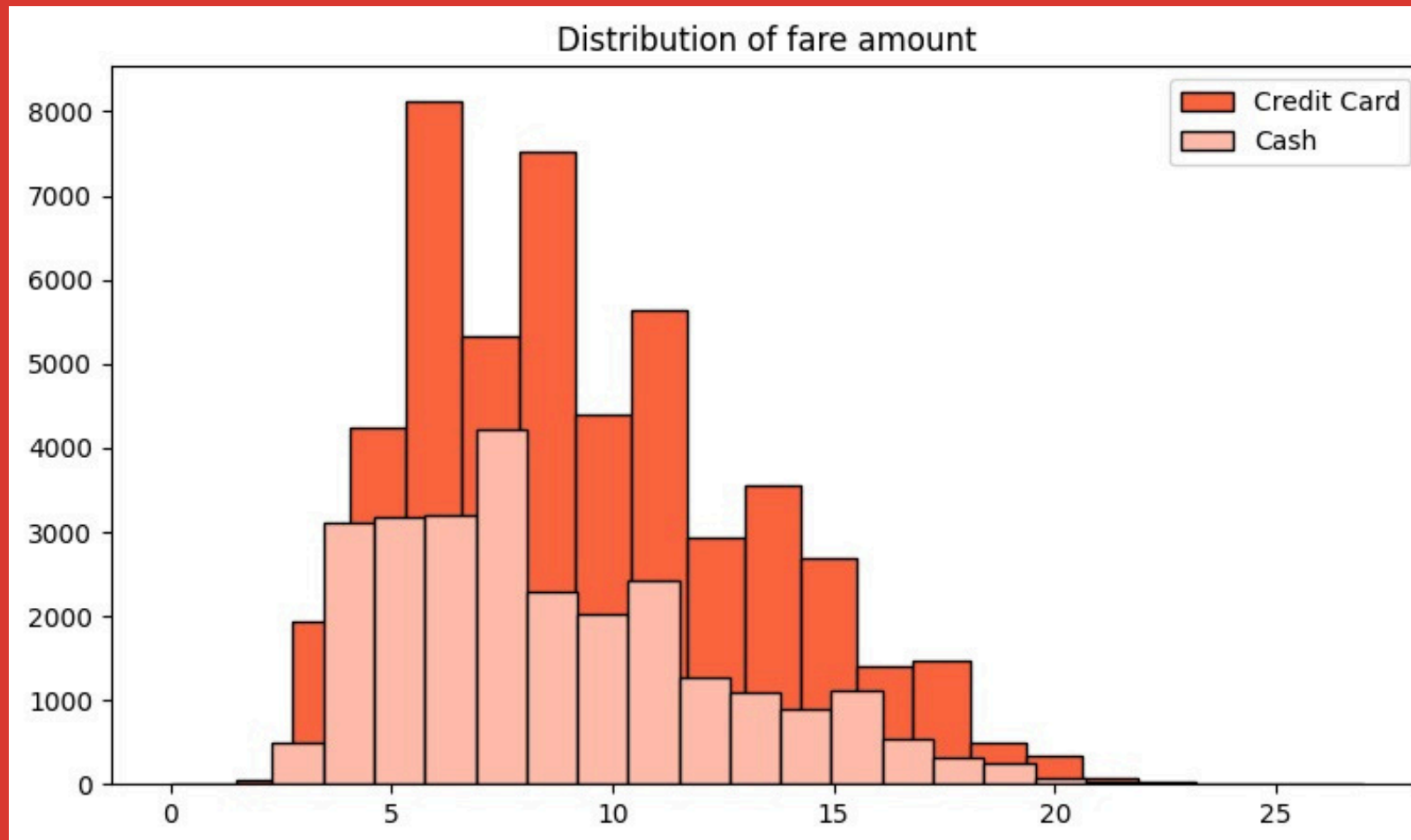
# Methodology

| Steps                  | Description                                                                                                                                                         |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Descriptive Statistics | Performed statistical analysis to summarize key aspects of the data, focusing on fare amounts and payment types.                                                    |
| Hypothesis Testing     | Conducted a T-test to evaluate the relationship between payment type and fare amount, testing the hypothesis that different payment methods influence fare amounts. |
| Regression Analysis    | Implemented linear regression to explore the relationship between trip duration (calculated from pickup and dropoff times) and fare amount.                         |



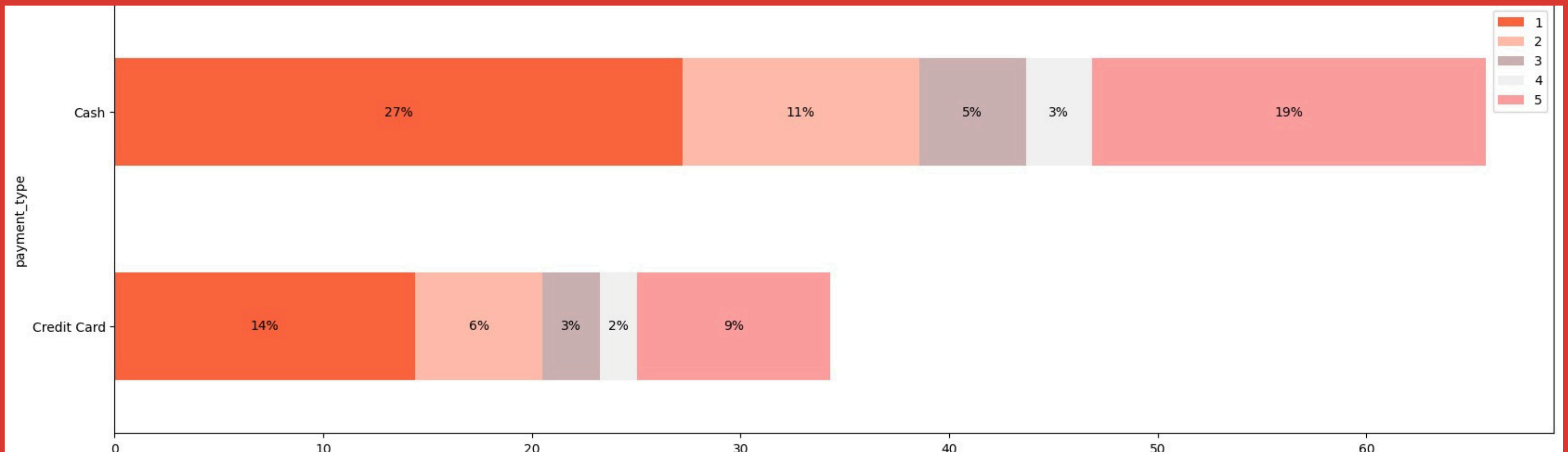
## Journey Insights

- Customers paying with cards tend to have a slightly higher average trip distance and fare amount compared to those paying with cash.
- Indicates that customers prefer to pay more with cards when they have high fare amount and long trip distance.



## Passenger Count Analysis

- Among card payments, rides with a single passenger (passenger\_count = 1) comprise the largest proportion, constituting 40.08% of all card transactions.
- Similarly, cash payments are predominantly associated with single-passenger rides, making up 20.04% of all cash transactions.
- There is a noticeable decrease in the percentage of transactions as the passenger count increases, suggesting that larger groups are less likely to use taxis or may opt for alternative payment methods.
- These insights emphasize the importance of considering both payment method and passenger count when analyzing transaction data, as they provide valuable insights into customer behavior and preferences.





# Hypothesis Testing

Null hypothesis: There is no difference in average fare between customers who use credit cards and customers who use cash.

Alternative hypothesis: There is a difference in average fare between customers who use credit cards and customers who use cash

With the T-statistic of 26.89 and a p-value of 0.00, we reject the null hypothesis and come to the conclusion that there is significant difference between the average fares for customers paying with credit card and cash.

# Recommendations



Encourage customers to pay with credit cards to capitalize on the potential for generating more revenue for taxi cab drivers.



Implement strategies such as offering incentives or discounts for credit card transactions to incentivize customers to choose this payment method.



**Thank You!!**